

Executive Summary

‘Clean energy’ has no single universal legal definition. For a carbon-focused answer, the defensible definition is low lifecycle greenhouse-gas emissions, measured in carbon-dioxide equivalent (CO₂e), not zero emissions at the point of operation. The IPCC uses a comparative ‘low-carbon’ concept: renewable sources, nuclear power, and fossil energy with carbon capture and storage (CCS) can all appear in low-carbon pathways, while unabated fossil fuels are excluded [1][2].

The clearest currently deployable low-carbon primary sources are wind, solar, hydropower, geothermal, marine energy where available, and nuclear fission. IPCC AR5 lifecycle medians are approximately 11 gCO₂e/kWh for onshore wind, 12 for offshore wind, 41–48 for photovoltaic solar, 24 for hydropower, 38 for geothermal, 27 for concentrating solar power, 17 for ocean energy, and 12 for nuclear; ranges can be wide because site, design, fuel supply and assumptions matter [3].

Biomass, waste-to-energy and fossil energy with CCS are conditional, not automatically clean. Hydrogen, ammonia, synthetic methane, methanol and e-kerosene are energy carriers, not primary energy sources; their carbon status depends on the electricity, feedstock and carbon source used to make them [2][4][5].

Introduction and Scope

This report answers ‘which energy sources are clean in terms of carbon emissions?’ in English, while retaining the distinctions used by the IPCC, ADEME, IEA, IRENA and NREL. ‘Lifecycle’ includes, as applicable, extraction or cultivation, materials and manufacturing, construction, fuel preparation, operation, maintenance, decommissioning and waste management. Lifecycle factors are normally expressed as gCO₂e/kWh of electricity; heat and hydrogen use different functional units.

The list is split into (A) low-carbon primary sources, (B) conditional or borderline pathways, (C) carriers and enabling technologies that are not sources, and (D) sources that are not clean on a lifecycle-carbon basis. The IPCC table below is an international literature synthesis, not a France-specific guarantee. ADEME’s Base Empreinte/Base Carbone values should be used for French projects because manufacturing mix, fuel cycle, reservoir conditions and system boundaries change results [6]. IRENA’s energy taxonomy is used to enumerate technology subtypes consistently [14], while NREL’s harmonised dataset is a cross-check on the spread of published lifecycle studies [15].

1. Low-carbon primary sources available today

Solar energy

Solar photovoltaic (PV) is a renewable primary source converted directly

into electricity. Include rooftop, utility-scale ground-mounted, floating, building-integrated, agrivoltaic and off-grid PV. IPCC AR5 gives lifecycle emissions of 26/41/60 gCO₂e/kWh for rooftop PV and 18/48/180 for utility PV (minimum/median/maximum) [3]. ADEME's current French public figures are roughly 23-44 gCO₂/kWh over the lifecycle, with older Base Carbone examples of 25.2 g/kWh for French-manufactured PV, 32.3 for a European manufacturing mix and 43.9 for the default Chinese mix [7]. PV is therefore low-carbon, but not impact-free: silicon, aluminium, glass, mining and manufacturing electricity dominate its footprint.

Concentrating solar power (CSP), also called solar-thermal electricity, uses mirrors and a receiver to produce heat and then electricity. Subtypes include parabolic trough, central-receiver/power-tower, linear Fresnel and dish/engine systems, often with molten-salt thermal storage. IPCC AR5 reports 8.8/27/63 gCO₂e/kWh [3]. CSP is commercially available but geographically constrained by strong direct sunlight; solar thermal heat for buildings, hot water, district heating and industrial process heat is also a low-carbon use of direct solar energy. ADEME reports a carbon impact of about 8 gCO₂/kWh for the collector alone and about 60 gCO₂/kWh when storage is included [8].

Wind energy

Onshore wind includes utility wind farms and small or micro-turbines. Offshore wind includes fixed-bottom and floating foundations; floating wind is newer but technically deployable. IPCC AR5 gives 7/11/56 gCO₂e/kWh for onshore and 8/12/35 for offshore wind [3]. ADEME's current French headline for onshore wind is about 12.7 gCO₂/kWh over the lifecycle [9]. Direct operating CO₂ is essentially zero; embodied emissions come mostly from steel, concrete, composites, vessels and grid connection. Airborne/kite wind is an emerging concept, not a mature mainstream source.

Hydropower

Run-of-river, diversion and conduit hydro use flowing water with little or no storage. Reservoir hydro uses dams and stored water. Pumped-storage hydro is primarily an electricity-storage technology, not a primary energy source; its carbon intensity depends on the electricity used to pump. IPCC AR5 gives hydropower 1/24/2200 gCO₂e/kWh. The very high upper bound reflects methane from some flooded reservoirs, particularly in warm, carbon-rich sites [3]. ADEME's current French public estimate is about 4-8 gCO₂/kWh lifecycle [10]. Hydro is generally low-carbon, but project-specific reservoir methane, land-use change and ecosystem impacts must be checked.

Geothermal energy

Conventional geothermal power includes dry-steam, flash-steam and binary-cycle plants. Direct geothermal heat supplies buildings, district heating, agriculture and industry. Enhanced geothermal systems

(EGS)/hot-dry-rock and closed-loop geothermal expand the resource base but remain emerging or demonstration-stage in many regions. IPCC AR5 reports 6/38/79 gCO₂/kWh for geothermal electricity [3]. IPCC AR6 distinguishes hydrothermal systems from EGS and shallow geothermal heat pumps [2]. Geothermal is renewable on human timescales, but drilling, construction, dissolved gases, reinjection and induced-seismicity management affect the result.

Marine and ocean energy

The complete marine-energy family includes tidal range (barrages and lagoons), tidal-stream/current turbines, wave energy, ocean-current energy, ocean thermal energy conversion (OTEC) and salinity-gradient/osmotic energy. IPCC AR5's quantified 'ocean' category covers tidal and wave technologies and reports 5.6/17/28 gCO₂/kWh [3]. OTEC, ocean-current and salinity-gradient systems have sparse, site-specific lifecycle evidence and are mostly pilot or early-commercial technologies; they should be described as low-carbon potential, not given a false universal number [2][11].

Nuclear fission

Nuclear is low-carbon but not renewable. The current family includes pressurised- and boiling-water reactors, heavy-water/pressurised-heavy-water reactors, gas-cooled reactors, fast-spectrum reactors, life extensions and new large reactors; small modular reactors are under development and early deployment. IPCC AR5 reports 3.7/12/110 gCO₂/kWh [3]. ADEME cites approximately 4 gCO₂/kWh for EDF's 2022 French lifecycle study, about 5 g in a CEA comparison, and a former Base Carbone factor of 6 g [6]. These figures include uranium mining, conversion, enrichment, fuel fabrication, plant construction/operation, decommissioning, reprocessing and waste management. They do not remove non-climate issues such as radioactive waste, accident risk, water use or mining impacts.

Ambient and recovered heat

Ambient renewable heat from air, surface water and shallow ground is harvested by air-source, water-source and ground-source heat pumps. The heat flow is renewable, but the pump's carbon intensity follows its electricity supply and refrigerant leakage. Waste-heat recovery from industry, wastewater, data centres and other processes is a low-carbon energy-recovery pathway, not a primary renewable source; its operating emissions can be near zero when the heat would otherwise be rejected [12].

2. Conditional or borderline low-carbon pathways

Bioenergy

Bioenergy includes solid biomass (wood, pellets, agricultural and forestry residues, bagasse, black liquor), biomass heat and power,

biogas/biomethane from anaerobic digestion or landfill gas, gasification/pyrolysis gases, and liquid biofuels such as ethanol, biodiesel, renewable diesel/HVO and sustainable aviation fuels. The renewable fraction of wastes can also be used.

The category is not automatically clean. IPCC AR5 gives 130/230/420 gCO₂e/kWh for dedicated biomass and 620/740/890 for biomass co-firing [3]. IPCC AR6 says lifecycle emissions can be incompatible with net zero depending on feedstock, land-use change, cultivation, transport, conversion efficiency and system boundary [2]. Residues, manure, sewage and unavoidable biogenic wastes can be much better than dedicated energy crops, but every pathway requires a feedstock-specific LCA. BECCS (bioenergy with CCS) is a carbon-removal configuration, not a universally carbon-negative source.

Waste-to-energy

Incineration, refuse-derived fuel, gasification, pyrolysis, anaerobic digestion and landfill-gas recovery can produce heat, electricity or fuels. Mixed municipal waste contains both biogenic and fossil carbon (plastics), and combustion also creates regulated air pollutants and residues. ADEME treats energy recovery as a last resort after prevention, reuse and recycling [13]. Classify waste-to-energy as conditional low-carbon, cleaner when it uses unavoidable biogenic residues and robust methane/air-pollution controls, not as intrinsically zero-carbon.

Fossil fuels with CCS/CCUS

Coal or gas with high-rate CO₂ capture, transport and permanent geological storage can be low-carbon in IPCC mitigation pathways, but it is not zero-carbon and CCS is a process, not a source. IPCC AR5 lifecycle values remain substantial: coal CCS roughly 100-250 gCO₂e/kWh and gas CCS roughly 94-340 gCO₂e/kWh, with residual combustion, upstream methane, energy penalty and possible storage leakage [3]. IPCC AR6 therefore requires minimal unabated fossil use and treats CCS in power as conditional and less mature [1][2].

3. Carriers and enabling technologies: clean only conditionally

Hydrogen is an energy carrier, not a primary energy source. Electrolytic green hydrogen can be low-carbon when powered by genuinely low-carbon renewable electricity; nuclear-powered electrolysis is low-carbon but not renewable. ADEME's current lifecycle figures per kg H₂ are approximately 0.45 (hydropower electrolysis), 0.70 (wind), 2.58 (PV), 2.77 (French grid) and 11.1 (natural-gas steam reforming) [4]. Blue hydrogen (gas reforming plus CCS) remains conditional because methane leakage and incomplete capture can erase much of the benefit [5].

Ammonia, e-methanol, e-methane, e-kerosene and other synthetic fuels are also carriers. They are low-carbon only if their hydrogen is low-carbon, their process electricity is low-carbon, and any carbon input comes from

biogenic sources or direct air capture rather than fossil CO₂.

Combustion that emits no fossil CO₂ at the point of use does not prove lifecycle carbon neutrality [2][4].

Batteries, pumped hydro, thermal storage, compressed air and flywheels store energy; they are not energy sources. Efficiency, demand reduction and smart-grid measures reduce emissions but are not sources either.

4. Not clean on a lifecycle-carbon basis

Unabated coal, oil and natural gas are not clean. IPCC AR5 medians are about 820 gCO₂e/kWh for coal and 490 for combined-cycle gas, against 11–48 for most wind, solar, hydro, geothermal, ocean and nuclear options [3]. Natural gas, LNG, renewable-looking gas without verified feedstock, and hydrogen made from unabated coal or gas must not be labelled clean.

Fusion has no commercial energy source today and no mature, agreed lifecycle factor. It belongs in future R&D, not in a list of currently available clean sources.

Synthesis and practical classification

For a strict present-day carbon screen, the 'clean' core is: solar PV, CSP, onshore and offshore wind, most well-sited hydropower, conventional geothermal, tidal and wave where deployed, and nuclear fission. Direct solar heat, geothermal heat, ambient heat pumps and genuinely recovered waste heat are low-carbon heat sources when electricity and equipment are included correctly.

The two most important caveats are variability and lifecycle boundaries. Wind and solar require grids, demand flexibility and storage; this does not make their generation carbon-intensive, but system-level emissions must be evaluated separately. Hydropower can be exceptionally low-carbon or unusually high-carbon if a reservoir produces methane. Biomass can be low-carbon when based on sustainable residues or wastes, but high-carbon when it drives land-use change. The correct comparison is therefore technology plus site, feedstock, electricity mix, lifetime and boundary—not a universal 'renewable = zero' label.

Limitations and Caveats

The IPCC table is a harmonised literature range, not a current plant certificate; ADEME's French factors use different vintages and boundaries. Values may change with manufacturing electricity, capacity factor, plant lifetime, reservoir conditions, fuel enrichment route, methane leakage and treatment of biogenic carbon. The numbers are CO₂-equivalent, not pure CO₂ only, and they do not cover biodiversity, land, water, toxicity, radioactive waste or social impacts.

Marine subtypes beyond tidal and wave, EGS, floating wind, advanced nuclear, blue hydrogen and synthetic fuels do not yet have one robust

global lifecycle factor. Use project-specific LCAs and disclose functional unit, system boundary, geography and whether storage/grid balancing is included.

Recommendations

1. Use 'low-lifecycle-GHG' rather than 'zero-carbon' in procurement and policy documents.
2. For France, start with ADEME Base Empreinte/Base Carbone factors and replace generic international medians with project-specific data.
3. Treat biomass, waste-to-energy and fossil CCS as conditional pathways requiring feedstock, methane, capture-rate and storage evidence.
4. Keep primary sources separate from carriers (hydrogen, ammonia, e-fuels) and storage (batteries, pumped hydro).
5. Report both plant-level lifecycle emissions and system-level impacts from grids, transmission, storage and backup.

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Methodology Appendix

The research triangulated approved IPCC AR5/AR6 reports, ADEME Base Empreinte/Base Carbone and current ADEME technology pages, plus IEA, IRENA, NREL and UNECE taxonomy/LCA material. Claims were separated into operational emissions, lifecycle emissions, and energy-carrier status. IPCC AR5 Table A.III.2 was used as the common numerical benchmark; ADEME French values were used where available. Emerging technologies were included only with an explicit maturity and uncertainty label.